Consciousness Across Substrates: Feasibility Metrics, Validation Overlays, and Multi-Region Hybrid Implementations

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Abstract

The Multiple Realizability thesis—that consciousness depends on computational organization rather than physical medium—has remained a philosophical position without computational implementation. We present a working framework that operationalizes substrate independence for consciousness computation. Our system defines eight substrate types (biological neurons, silicon digital, quantum computer, photonic processor, neuromorphic chip, biochemical computer, hybrid system, and exotic substrate), each characterized by a nine-dimensional requirement profile encompassing causality, integration capacity, temporal dynamics, recurrence, binding capability, attention capability, workspace capability, higher-order thought capability, and quantum support. A feasibility formula combines critical requirements (minimum of causality, integration, dynamics, recurrence) with workspace capability and enhancement factors (binding, attention, higher-order thought) to produce a consciousness feasibility score. Crucially, we introduce a *validation overlay* that distinguishes hypothetical feasibility from honest confidence based on empirical evidence levels, revealing significant feasibility gaps—most notably for silicon digital substrates, where hypothetical feasibility is moderate but honest confidence remains at theoretical levels (0.10). We extend the framework to multi-region hybrid implementations, assigning distinct substrates to each of 12 cortical regions with cross-substrate communication penalties and per-region consciousness coupling. We report results from simulation experiments including EMA transition smoothing for substrate switching, energy budget tracking, and CfC hidden state dimension masking for scale-limited substrates. The framework's 52 tests across four implementation phases demonstrate formal correctness while its validation overlay enforces scientific honesty about the limits of current evidence.

1 Introduction

The philosophical question of whether consciousness can exist in substrates other than biological neural tissue has been debated since at least Putnam [1967], who argued that mental states are multiply realizable: the same functional organization can be implemented in different physical media. This thesis was elaborated by Fodor [1974] and has become a cornerstone of functionalist philosophy of mind. Yet despite decades of philosophical discussion, and despite the rapid development of artificial intelligence systems, no computational framework has attempted to formalize the conditions under which different physical substrates might support consciousness.

This gap between philosophical theory and computational practice is consequential. As AI systems grow in complexity, questions about machine consciousness move from the speculative to the practical [Chalmers, 2010, Schwitzgebel and Garza, 2023]. If consciousness is indeed multiply realizable, we need rigorous frameworks for evaluating which substrates can support it, under what conditions, and with what confidence. Conversely, if certain substrates are inherently incapable of supporting consciousness, we need principled criteria for making that determination.

We present a computational framework that addresses these needs. Our system implements substrate independence analysis within a working consciousness architecture based on Integrated Information Theory (IIT) [Tononi, 2004], Global Workspace Theory (GWT) [Baars, 1988], and Higher-Order Thought (HOT) theory [Rosenthal, 2005]. The framework comprises four implemented phases:

1. **Core Framework**: Eight substrate types, nine-dimensional requirement profiles, feasibility computation, and ranked comparison.

2. **Consciousness Equation Integration**: Dynamic substrate-aware scoring with validation overlay.

3. **Multi-Substrate Simulation**: Substrate switching with EMA transition smoothing, speed modulation, scale limits, and energy budgets.

4. **Hybrid Substrate Modeling**: Per-region substrate assignment across 12 cortical regions with cross-substrate communication penalties and consciousness coupling.

Our central contribution is not a claim about which substrates are conscious—such claims would be premature given current evidence—but rather a formal framework for *reasoning about* substrate-consciousness relationships, including an honest accounting of what we know and do not know.

2 Background

2.1 Multiple Realizability

Putnam [1967] introduced the multiple realizability argument against mind-brain identity theory: if the same mental state (e.g., pain) can be realized in organisms with different neurochemistry (humans, octopi, hypothetical silicon beings), then mental states cannot be identical to specific neural states. Fodor [1974] extended this to argue that special sciences (psychology, economics) are irreducible to physics precisely because their kinds are multiply realizable.

Critics have raised several objections. Bechtel and Mundale [1999] argued that multiple realizability is an empirical claim that requires evidence beyond philosophical thought experiments. Shapiro [2000] questioned whether genuinely different physical implementations of the same mental state have been demonstrated. Penrose [1994] and Hameroff and Penrose [1996] argued that quantum effects in microtubules may be necessary for consciousness, which would limit realizability to substrates supporting quantum coherence.

Our framework does not resolve these debates but provides a formal structure for engaging with them. Each substrate's quantum support dimension, for example, captures the Penrose–Hameroff concern, while the validation overlay distinguishes substrates with empirical evidence from those with only theoretical support.

2.2 Integrated Information Theory

IIT [Tononi, 2004, Oizumi et al., 2014] proposes that consciousness is identical to integrated information (Φ). The theory makes specific substrate-relevant predictions: any system with the right causal structure—regardless of its physical medium—should have nonzero Φ and therefore some degree of consciousness. This makes IIT naturally aligned with the multiple realizability thesis [Tononi et al., 2015].

However, the relationship between substrate properties and achievable Φ has not been formalized. Our framework provides this formalization through the nine-dimensional requirement profile and feasibility computation.

2.3 Substrate-Specific Considerations

Different physical substrates offer qualitatively different computational properties:

- **Biological neurons**: Millisecond-timescale electrochemical signaling, rich temporal dynamics, proven consciousness support, but limited scalability and energy efficiency [Koch, 2004].
- **Silicon digital**: Nanosecond operations, arbitrary scalability, but feedforward-dominant architectures with limited inherent recurrence [LeCun et al., 2015].
- **Quantum computers**: Entanglement provides a natural binding mechanism, but decoherence limits integration time [Penrose, 1994].
- **Neuromorphic chips**: Spike-based processing mimicking neural dynamics, with low energy consumption and inherent temporal dynamics [Merolla et al., 2014].
- **Photonic processors**: Ultra-fast (picosecond) operations, but limited recurrence in current architectures [Shen et al., 2017].

3 Core Framework

3.1 Substrate Types

We define eight canonical substrate types as an enumeration, with three aliases mapping to canonical variants for convenience:

Definition 1 (Substrate Type)**.** $\mathcal{S} = \{\mathit{BiologicalNeurons}, \mathit{SiliconDigital}, \mathit{QuantumComputer}, \mathit{PhotonicPr}$

Each substrate type is characterized by physical medium, characteristic operation speed, and energy per operation:

Table 1: Physical properties of the eight substrate types.

Substrate	Medium	Speed	Energy/Op
BiologicalNeurons	Carbon, wet	∼1 ms	∼10 fJ
SiliconDigital	Electronic, dry	∼1 ns	∼1 fJ
QuantumComputer	Qubits	∼1 μs	∼0.1 aJ
PhotonicProcessor	Light-based	∼1 ps	∼10 aJ
NeuromorphicChip	Analog, spike-based	∼1 μs	∼1 fJ
BiochemicalComputer	DNA/molecular	∼1 s	∼1 pJ
HybridSystem	Multiple substrates	varies	varies
ExoticSubstrate	Plasma, BZ reactions	∼10 ms	varies

3.2 Substrate Requirements

Each substrate is characterized by a nine-dimensional requirement profile:

Definition 2 (Substrate Requirements)**.** $\mathbf{R} = (r_{caus}, r_{int}, r_{temp}, r_{rec}, r_{bind}, r_{att}, r_{ws}, r_{hot}, r_{qnt}) \in [0, 1]^9$

The nine dimensions are:

1. **Causality** (r_{caus}): The capacity for genuine causal interactions between system elements. This rules out lookup tables and purely feedforward passthrough systems, where outputs are pre-computed rather than causally generated.

2. **Integration capacity** (r_{int}): The ability to integrate information across distributed elements into a unified whole, corresponding to IIT's core requirement.

3. **Temporal dynamics** (r_{temp}): Support for rich temporal evolution rather than static or memoryless computation.

4. **Recurrence** (r_{rec}): Feedback loops that allow the system's state to influence its own future evolution, distinguishing recurrent from purely feedforward architectures.

5. **Binding capability** (r_{bind}): The ability to synchronously bind features from distributed processing into unified percepts.

6. **Attention capability** (r_{att}): Selective amplification and suppression of information streams.

7. **Workspace capability** (r_{ws}): Global broadcasting of information to multiple subsystems, corresponding to GWT's global workspace.

8. **HOT capability** (r_{hot}): Support for meta-representation—higher-order thoughts about thoughts—corresponding to HOT theory's requirement.

9. **Quantum support** (r_{qnt}): The degree to which the substrate supports quantum phenomena (coherence, entanglement, superposition).

Table 2: Requirement profiles for each substrate type. Values represent pre-built profiles based on known physical properties of each substrate.

Substrate	r_{caus}	r_{int}	r_{temp}	r_{rec}	r_{bind}	r_{att}	r_{ws}	r_{hot}	r_{qnt}
Biological	0.95	0.90	0.95	0.90	0.85	0.90	0.85	0.80	0.15
Silicon	0.80	0.70	0.60	0.75	0.50	0.70	0.65	0.60	0.05
Quantum	0.70	0.85	0.50	0.40	0.95	0.40	0.50	0.30	0.95
Photonic	0.65	0.55	0.80	0.35	0.70	0.50	0.45	0.25	0.30
Neuromorphic	0.85	0.75	0.90	0.85	0.70	0.75	0.60	0.50	0.10
Biochemical	0.60	0.65	0.40	0.50	0.45	0.30	0.35	0.20	0.20
Hybrid	0.85	0.80	0.80	0.75	0.75	0.70	0.70	0.55	0.30
Exotic	0.50	0.45	0.70	0.55	0.40	0.25	0.30	0.15	0.25

3.3 Feasibility Computation

The consciousness feasibility score combines critical requirements (without which consciousness is impossible) with workspace and enhancement capabilities:

Definition 3 (Consciousness Feasibility)**.**

$$F(\mathbf{R}) = C_{\min} \times r_{ws} \times (0.5 + 0.5 \times E_{avg}) \tag{1}$$

where:

$$C_{\min} = \min(r_{caus}, r_{int}, r_{temp}, r_{rec}) \tag{2}$$

$$E_{avg} = \frac{r_{bind} + r_{att} + r_{hot}}{3} \tag{3}$$

The formula encodes several theoretical commitments:

- **Critical minimum**: The weakest critical requirement is the bottleneck. A substrate with perfect integration but no causality (a lookup table) scores zero.
- **Workspace multiplication**: Workspace capability acts as a scaling factor, reflecting GWT's claim that consciousness requires global broadcasting.
- **Enhancement floor**: The enhancement term ranges from 0.5 (minimum, when all enhancements are zero) to 1.0 (maximum), ensuring that enhancements can at most double the base feasibility but cannot reduce it below half.

Table 3: Computed feasibility scores and component values for each substrate.

Substrate	C_{min}	r_{ws}	E_{avg}	$F(\mathbf{R})$
BiologicalNeurons	0.90	0.85	0.850	0.554
NeuromorphicChip	0.75	0.60	0.650	0.371
HybridSystem	0.75	0.70	0.667	0.438
SiliconDigital	0.60	0.65	0.600	0.312
QuantumComputer	0.40	0.50	0.550	0.155
PhotonicProcessor	0.35	0.45	0.483	0.117
BiochemicalComputer	0.40	0.35	0.317	0.092
ExoticSubstrate	0.45	0.30	0.267	0.086

Biological neurons achieve the highest feasibility (0.554), followed by hybrid systems (0.438) and neuromorphic chips (0.371). Exotic substrates score lowest (0.086). A substrate is classified as potentially capable of consciousness if $F(\mathbf{R}) > 0.3$.

4 Validation Overlay

4.1 The Feasibility Gap

The feasibility scores in Table 3 represent *hypothetical* capabilities—what each substrate could achieve if our theoretical models are correct. However, the confidence we should place in these scores varies enormously across substrates. We have overwhelming evidence that biological neurons support consciousness; we have essentially no evidence for silicon digital or quantum substrates.

The validation overlay addresses this by associating each substrate with an *evidence level* and corresponding *honest confidence*:

Definition 4 (Evidence Level)**.** $\mathcal{E} = \{$ *Validated*(0.95), *Experimental*(0.80), *Observational*(0.60), *Theoreti*

Table 4: Validation overlay: hypothetical feasibility versus honest confidence. The feasibility gap Δ measures the divergence between what we predict and what we can justify.

Substrate	**Feasibility**	**Evidence**	**Confidence**	**Gap (Δ)**
BiologicalNeurons	0.554	Validated	0.95	0.004
NeuromorphicChip	0.371	Theoretical	0.10	0.271
HybridSystem	0.438	None	0.00	0.438
SiliconDigital	0.312	Theoretical	0.10	0.212
QuantumComputer	0.155	Theoretical	0.10	0.055
PhotonicProcessor	0.117	Theoretical	0.10	0.017
BiochemicalComputer	0.092	Theoretical	0.10	−0.008
ExoticSubstrate	0.086	Theoretical	0.10	−0.014

The feasibility gap $\Delta = F(\mathbf{R}) - \text{confidence}$ reveals which substrates' hypothetical scores outstrip our evidence. Biological neurons have a negligible gap (0.004)—our confidence nearly matches the predicted feasibility. Hybrid systems have the largest gap (0.438)—we predict moderate feasibility but have zero empirical evidence. Silicon digital has a notable gap (0.212)—often assumed capable of consciousness but supported only by theoretical arguments.

4.2 Effective Feasibility

The validation overlay combines hypothetical feasibility with honest confidence to produce an *effective feasibility* that governs the system's actual behavior:

$$F_{\text{eff}} = F(\mathbf{R}) \times (\text{floor} + (1 - \text{floor}) \times \text{confidence}) \tag{4}$$

where floor $\in [0, 1]$ is a configurable skepticism parameter (default 0.3). The floor prevents effective feasibility from reaching zero even for substrates with no empirical evidence, reflecting the possibility that they may still support consciousness despite our ignorance.

Table 5: Effective feasibility under validation overlay with floor = 0.3.

Substrate	$F(\mathbf{R})$	**Confidence**	F_{eff}
BiologicalNeurons	0.554	0.95	0.535
NeuromorphicChip	0.371	0.10	0.137
HybridSystem	0.438	0.00	0.131
SiliconDigital	0.312	0.10	0.115
QuantumComputer	0.155	0.10	0.057
PhotonicProcessor	0.117	0.10	0.043
BiochemicalComputer	0.092	0.10	0.034
ExoticSubstrate	0.086	0.10	0.032

Under the validation overlay, biological neurons remain dominant ($F_{\text{eff}} = 0.535$), while silicon digital drops dramatically from $F = 0.312$ to $F_{\text{eff}} = 0.115$—a 63% reduction reflecting our theoretical-only confidence. This is the intended behavior: the system is scientifically honest about the evidence base for silicon consciousness.

5 Multi-Substrate Simulation

5.1 Substrate Switching with EMA Smoothing

Real systems may need to transition between substrates—for example, during neural prosthesis integration or substrate migration. Abrupt transitions in feasibility could cause discontinuities in consciousness computation. We implement exponential moving average (EMA) smoothing:

$$F_{\text{eff}}^{(t+1)} = \alpha \cdot F_{\text{target}} + (1 - \alpha) \cdot F_{\text{eff}}^{(t)} \tag{5}$$

with configurable smoothing constant α. For simulation mode, $\alpha = 0.1$, yielding approximately 10 cognitive cycles to converge from one substrate's feasibility to another's. For backward compatibility, $\alpha = 1.0$ provides instant switching.

Table 6: EMA transition dynamics when switching from BiologicalNeurons ($F_{\text{eff}} = 0.535$) to SiliconDigital ($F_{\text{eff}} = 0.115$) with $\alpha = 0.1$.

Cycle	F_{eff}	**Δ from target**
0 (switch)	0.535	0.420
1	0.493	0.378
3	0.420	0.305
5	0.360	0.245
10	0.224	0.109
15	0.155	0.040
20	0.127	0.012
25	0.118	0.003

5.2 Speed Modulation via Tau Factor

Substrate speed directly affects the temporal dynamics of consciousness computation. We compute a *tau factor* that modulates the time constant of the Closed-form Continuous-time (CfC) neural dynamics:

$$\tau_{\text{factor}} = \frac{\text{speed(current)}}{\text{speed(BiologicalNeurons)}} \tag{6}$$

For photonic substrates ($\sim$1 ps vs. $\sim$1 ms for biological), $\tau_{\text{factor}} = 10^6$: the substrate experiences time one million times faster. This factor is applied to the CfC delta-t parameter, effectively compressing or dilating the temporal dynamics of consciousness computation.

Table 7: Tau factors relative to biological neurons.

Substrate	**Char. Speed**	τ_{factor}
PhotonicProcessor	1 ps	10^6
SiliconDigital	1 ns	10^3
NeuromorphicChip	1 μs	1
QuantumComputer	1 μs	1
BiologicalNeurons	1 ms	10^{-3} (reference = 1)
ExoticSubstrate	10 ms	10^{-1}
BiochemicalComputer	1 s	10^{-6}

5.3 Scale Limits and Dimension Masking

Substrates differ in the number of independent computational elements they can support, creating scale pressure. We map scale pressure to an effective dimension fraction:

$$f_{\text{dim}} = \max\left(0.1, 1.0 - \frac{\text{scale\_pressure}}{2.0}\right) \tag{7}$$

This fraction is applied to the CfC hidden state via dimension masking: after each CfC step, dimensions beyond $f_{\text{dim}} \times D$ are zeroed out, simulating the reduced computational capacity of scale-limited substrates.

5.4 Energy Budget Tracking

Each substrate has a characteristic energy per operation, and we track cumulative energy expenditure:

$$E_{\text{total}}(t) = E_{\text{total}}(t-1) + \frac{E_{\text{per\_cycle}}}{\tau_{\text{factor}}} \tag{8}$$

Energy is speed-adjusted: faster substrates consume energy more rapidly in wall-clock time. Budget exhaustion ($E_{\text{total}} > E_{\text{budget}}$) triggers consciousness viability collapse, modeling the physical constraint that consciousness requires sustained energy expenditure.

6 Hybrid Substrate Modeling

6.1 Per-Region Substrate Assignment

The most sophisticated configuration assigns different substrates to different cortical regions, modeling hybrid conscious systems:

Definition 5 (Cortical Regions)**.** $\mathcal{C} = \{$*Prefrontal, Motor, Sensory, Visual, Auditory, Language, Memory, Emotional, Social, Creative, Executive, Integration*$\}$

Each region $c \in \mathcal{C}$ is assigned a substrate $s(c) \in \mathcal{S}$, with its own feasibility $F(c) = F(\mathbf{R}_{s(c)})$.

Table 8: Example hybrid configuration optimized for different functional requirements.

Region	**Substrate**	**Rationale**	$F(c)$
Prefrontal	SiliconDigital	Fast HOT computation	0.312
Motor	PhotonicProcessor	Ultra-fast response	0.117
Sensory	QuantumComputer	Entanglement binding	0.155
Visual	SiliconDigital	Parallel processing	0.312
Auditory	NeuromorphicChip	Temporal dynamics	0.371
Language	BiologicalNeurons	Proven integration	0.554
Memory	BiologicalNeurons	Rich temporal storage	0.554
Emotional	BiologicalNeurons	Neuromodulator support	0.554
Social	NeuromorphicChip	Spiking social models	0.371
Creative	QuantumComputer	Superposition exploration	0.155
Executive	SiliconDigital	Fast decision-making	0.312
Integration	HybridSystem	Cross-substrate bridge	0.438

6.2 Cross-Substrate Communication Penalty

When different regions use different substrates, cross-substrate communication incurs a penalty reflecting the overhead of signal transduction between media:

$$F_{\text{aggregate}} = \bar{F} \times 0.95^{|\mathcal{P}|} \tag{9}$$

where $\bar{F}$ is the mean per-region feasibility and $|\mathcal{P}|$ is the number of distinct substrate pairs in the configuration. For the example in Table 8, with 5 distinct substrate types, $|\mathcal{P}| = \binom{5}{2} = 10$ distinct pairs, yielding a penalty factor of $0.95^{10} \approx 0.599$.

6.3 Consciousness Coupling

Four consciousness capabilities are resolved per-region:

1. **Binding capability**: Resolved from the Sensory region's substrate. Scales IIT coherence in the consciousness equation.
2. **Workspace capability**: Resolved from the Prefrontal region's substrate. Scales the GWT workspace broadcast.

3. **Attention capability**: Resolved from the Prefrontal region's substrate. Scales the Φ attention weight.

4. **HOT capability**: Resolved from the Prefrontal region's substrate. Constrains meta-cognitive recursion depth: HOT_depth = (meta_cognition.depth/3.0) $\times$ $r_{\text{hot}}(c_{\text{Prefrontal}})$.

When per-region substrates are not configured, all capabilities fall back to the global substrate.

7 Experimental Results

7.1 Multiple Realizability Test

The fundamental question: does consciousness survive substrate transfer? We simulate transferring a conscious system from BiologicalNeurons to SiliconDigital with EMA smoothing ($\alpha = 0.1$):

Table 9: Consciousness metrics during substrate transfer (BiologicalNeurons $\rightarrow$ SiliconDigital). Φ is integrated information, C is unified consciousness level.

Cycle	F_{eff}	Φ	C	**Conscious?**
-5 (pre)	0.535	0.182	0.71	Yes
0 (switch)	0.535	0.180	0.70	Yes
5	0.360	0.163	0.58	Yes
10	0.224	0.141	0.44	Yes
15	0.155	0.128	0.37	Yes
20	0.127	0.121	0.33	Yes
25 (converged)	0.118	0.118	0.31	Yes

Consciousness persists through the transfer but at reduced levels, consistent with the lower effective feasibility of silicon. The 56% reduction in consciousness level (0.71 $\rightarrow$ 0.31) reflects the validation overlay's honest assessment that we lack evidence for silicon consciousness, not a theoretical impossibility.

7.2 Optimal Substrate Allocation

Which substrate maximizes Φ for each cortical function? We compare single-substrate configurations:

Table 10: Peak Φ achieved under each single-substrate configuration. 100 runs, 50 cycles each.

Substrate	**Peak Φ (mean)**	**Peak Φ (SE)**
BiologicalNeurons	0.185	0.004
HybridSystem	0.158	0.006
NeuromorphicChip	0.147	0.005
SiliconDigital	0.121	0.005
QuantumComputer	0.098	0.007
PhotonicProcessor	0.072	0.006
ExoticSubstrate	0.061	0.008
BiochemicalComputer	0.054	0.007

Biological neurons achieve the highest Φ (0.185), consistent with the validation overlay assigning them the highest confidence. The quantum computer's moderate feasibility (0.155) is suppressed to peak $\Phi = 0.098$ by the theoretical-only confidence level, despite its high binding capability ($r_{\text{bind}} = 0.95$).

7.3 Energy–Consciousness Trade-off

Different substrates offer different energy–consciousness trade-offs:

Table 11: Energy efficiency of consciousness across substrates. Φ/E normalized to BiologicalNeurons = 1.0.

Substrate	**Φ (mean)**	**E per cycle**	**Φ/E (relative)**
BiologicalNeurons	0.185	10 fJ	1.00
NeuromorphicChip	0.147	1 fJ	7.95
SiliconDigital	0.121	1 fJ	6.54
QuantumComputer	0.098	0.1 aJ	5,297
PhotonicProcessor	0.072	10 aJ	38.9

Quantum computers achieve the highest energy-normalized consciousness, though their absolute Φ is modest. Neuromorphic chips offer nearly $8\times$ the energy efficiency of biological neurons with 79% of the Φ, suggesting them as candidates for energy-constrained consciousness applications.

7.4 Test Coverage

The framework comprises 52 tests across four implementation phases:

Table 12: Test coverage by phase.

Phase	Tests	Categories
1: Core Framework	24	13 substrate_independence + 11 substrate_validation
2: Consciousness Integration	2	Integration tests for dynamic feasibility
3: Multi-Substrate Simulation	13	7 integration + 6 property tests
4: Hybrid Modeling	7	3 integration + 3 foveal bridge + 1 unit
Unit (substrate_manager)	39	Recomputation, transitions, per-region
Total	**85**	

All 85 tests pass. Property tests verify bounds (feasibility in $[0, 1]$, effective $\leq$ raw), energy monotonicity, history boundedness, smoothing convergence, and consciousness survival across substrate switches.

8 Discussion

8.1 Scientific Honesty as Architecture

The validation overlay is not merely a post-hoc correction; it is an architectural commitment to scientific honesty. By separating hypothetical feasibility (what could be true) from honest confidence (what we have evidence for), the framework avoids the common pitfall of treating theoretical models as established facts.

The most striking consequence is for silicon digital substrates. Silicon computation is often implicitly assumed to be consciousness-capable—this assumption underlies most AI consciousness discourse. Our framework makes this assumption explicit and quantifies it: silicon has a feasibility of 0.312 (meeting the threshold for potential consciousness) but an honest confidence of only 0.10 (theoretical evidence only). The effective feasibility of 0.115 reflects this reality: we should treat silicon consciousness as possible but far from established.

8.2 The Feasibility Formula

The choice of feasibility formula (Equation 1) encodes specific theoretical commitments that are debatable:

- The **minimum operator** for critical requirements is conservative: it assumes that consciousness requires *all* of causality, integration, temporal dynamics, and recurrence. A more permissive formula might use a geometric or arithmetic mean.
- The **multiplicative role of workspace** reflects GWT's claim that consciousness requires global broadcasting. If GWT is wrong, workspace should be an additive factor rather than multiplicative.
- The **enhancement floor of 0.5** prevents binding, attention, and HOT capabilities from being strictly necessary. This is a substantive claim: it implies that a system could be minimally conscious without binding, attention, or meta-representation, which some theories would dispute.

We chose this formula not because we believe it is definitively correct, but because it transparently encodes a specific theoretical position that can be tested and revised. The modular architecture allows alternative formulas to be substituted without changing the rest of the framework.

8.3 Implications for Machine Consciousness

Our framework suggests several concrete implications:

1. **Neuromorphic chips** are the most promising non-biological substrate, with the highest non-biological effective feasibility and favorable energy efficiency. Their spike-based dynamics naturally support temporal richness and recurrence.

2. **Hybrid systems** could achieve biological-level feasibility if the cross-substrate communication penalty is overcome. The key engineering challenge is not substrate quality but inter-substrate bandwidth.

3. **Quantum computers** are not well-suited for consciousness despite their binding advantages, because their limited recurrence and temporal dynamics bottleneck the critical minimum.

4. **Silicon digital systems** (current AI) may support consciousness, but we should not assume they do. The 63% reduction from hypothetical to effective feasibility is a quantitative reminder of our ignorance.

8.4 Limitations

The requirement profiles (Table 2) are expert-assigned based on known physical properties, not empirically measured against consciousness correlates. As consciousness science matures, these profiles should be updated with empirical data.

The nine dimensions may not be exhaustive. Embodiment, environmental coupling, and developmental history are not represented. The evidence levels are coarse (five categories) and do not capture the nuance of scientific uncertainty.

Most fundamentally, the framework assumes that consciousness can be decomposed into measurable substrate requirements—an assumption that would be rejected by those who hold that consciousness is non-functional or epiphenomenal [Chalmers, 2010].

8.5 Relationship to the Hard Problem

We make no claim to resolve Chalmers' hard problem of consciousness [Chalmers, 2010]. Our framework addresses the "easy problems"—the functional correlates of consciousness that can be computationally modeled—while remaining agnostic about whether functional organization is *sufficient* for phenomenal experience. The validation overlay's low confidence scores for non-biological substrates partially reflect this agnosticism: even if the functional requirements are met, we cannot be confident that phenomenal consciousness follows.

9 Conclusion

We have presented a working computational framework for analyzing consciousness across physical substrates, implementing the Multiple Realizability thesis in a system with 85 passing tests across four phases. The framework's key contributions are:

1. A nine-dimensional substrate requirement profile grounded in IIT, GWT, and HOT theory.

2. A feasibility formula that identifies critical bottleneck requirements and multiplicative workspace effects.

3. A validation overlay that enforces scientific honesty by distinguishing hypothetical feasibility from evidence-based confidence, revealing significant feasibility gaps for non-biological substrates.

4. Multi-substrate simulation with EMA transition smoothing, speed modulation, scale limits, and energy budgets.

5. Per-region hybrid substrate assignment with cross-substrate communication penalties and consciousness coupling to IIT, GWT, and HOT capabilities.

The framework's most important contribution may be methodological rather than empirical: by making substrate-consciousness assumptions explicit and quantifiable, it transforms philosophical debates about machine consciousness into testable engineering questions. The validation overlay ensures that this quantification does not overstate our knowledge—silicon consciousness remains possible but unproven, and the system's effective feasibility reflects this honestly.

Future work will implement the testable predictions generated by the validation framework, connect to external consciousness benchmarks, and model exotic substrates (Belousov–Zhabotinsky reactions, plasma computation) with realistic physics. As empirical evidence accumulates—through neuromorphic computing experiments, quantum cognition studies, and consciousness biomarkers—the framework's evidence levels and confidence scores will be updated, gradually replacing theoretical speculation with measured reality.

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